Unix and Bash exercises

NUnix · L3 MIAGE · TD

ISTIC · University of Rennes 2026–2027

Université de Rennes
Louis Ledoux

Exercise session

Université de Rennes
Louis Ledoux

1. Filename patterns
2. Redirections and paths
3. Scripts with arguments and interactive input
4. Search commands

Use a **practice directory** for files that you create or change.
Read commands and predict their effects before executing them.

Exercise 1: filename patterns

Université de Rennes
Louis Ledoux

List the entries in `/usr/bin` whose names:

1. Start with `m`.
2. Start with `m` and have exactly three characters.
3. Start with `m` and have at least three characters.
4. Start with `m` and have an extension after a dot.

Exercise 2: the initial files

Université de Rennes
Louis Ledoux

Directory	Contains
`practice/user1/`	`f1`, `f2`, `f3`
`practice/user2/`	`f4`

The current directory is **`practice/user2/`**.

What relative path names `user1`'s file `f1`?

Exercise 2: redirections with cat

Université de Rennes
Louis Ledoux

Using `cat` and redirections:

1. Create `f5` containing the line `5 2 3 -1`.
2. Create `f6` from `f5`, followed by `f4`.
3. Append `user1`'s `f1` to `f6`.
4. Copy `user1`'s `f2` into a new file `f7` in `user2`.

Exercise 3: a script with three filenames

Université de Rennes
Louis Ledoux

Write a script named `execution`, called with `n1 n2 n3`.

Step	Input	Program	Output
1	File `n1`	`p1`	File `n1.r1`
2	File `n2`	`p1`	File `n2.r1`
3	`n2.r1`, then `n1.r1`	`p2`	File `n3`

`p1` and `p2` read **stdin** and write **stdout**.
Concatenate the two intermediate files in the stated order.

Exercise 3: resolving the paths

Université de Rennes
Louis Ledoux

Directory	Contains
`practice/user1/`	`f1`, `f2`, `f3`, `p1`, `p2`, `execution`
`practice/user2/`	`f4`

From `practice/user1/`, run:

```
./execution f1 ../user2/f4 f5
```

Which files are read? Where do the three output files appear?

7/13

Exercise 3: interactive input

Université de Rennes
Louis Ledoux

Write another version of `execution` that asks the user for:

- The first input filename.
- The second input filename.
- The output filename.

Keep the same processing steps and concatenation order.

8/13

Exercise 4: script arguments

Université de Rennes
Louis Ledoux

Write `argument`, using `expr` to update a counter.

```
./argument one two three
```

Expected output:

```
3 arguments:
 1 - one
 2 - two
 3 - three
```

9/13

Exercise 5: finding a pattern

Université de Rennes
Louis Ledoux

Write `mysearch`, called with a pattern and a filename:

```
./mysearch 'error' application.log
```

Print either:

- `application.log has the pattern`
- `application.log does not have the pattern`

Handle an unreadable or missing file as an **error**.

10/13

Exercise 5: repeated searches

Université de Rennes
Louis Ledoux

1. Write `mysearch2 filename`. Ask for a pattern repeatedly until the user enters **end**.
2. Write `mysearch3 f1 ... fn`. Search the supplied files for each entered pattern.

Where does each loop belong? What happens when input reaches EOF?

11/13

Exercise 6: reading find commands

Université de Rennes
Louis Ledoux

Explain each command. Identify unnecessary directory visits or ambiguous patterns.

```
find /var/log -exec grep 'www.univ-rennes1.fr' {} \;
find . -name 'rc.conf' -exec chmod o+r {} \;
find /usr/local -type f -name '*.html'
```

12/13

Exercise 6: who handles the redirection?

Université de Rennes
Louis Ledoux

Analyse this command before running it:

```
find . -name 'tp2' -exec {} < f_in > {}.out \;
```

1. Which program opens `f_in` and `{}.out`?
2. Does `find` replace `{}` inside the redirection filename?
3. How would you give each matching program its own output file?

13/13